

Noah Horne

MSc Mathematical & Theoretical Physics, University of Oxford

✉ noah@horne.ws

🌐 [linkedin.com/in/noahhorne](https://www.linkedin.com/in/noahhorne)

🌐 noahhorne.com



Education

University of Oxford

MSc Mathematical & Theoretical Physics

Graduating Jul 2027

Oxford, United Kingdom

Constructor University

BSc Physics & Data Science

Graduated Jun 2026

Bremen, Germany

- **Final GPA: 1.07** (German scale, 1.0 = highest; UK equivalent: First-Class Honours)
- Awarded the maximum merit-based scholarship (approx. €24,000)

Anglo-American School of Sofia

International Baccalaureate Diploma

Graduated May 2023

Sofia, Bulgaria

- **Final score: 44/45** (UK equivalent: A*)
- Valedictorian of the graduating class

Research Experience & Technical Projects

Summer University in Plasma Physics

Max Planck Institute for Plasma Physics

Sep 2026

Germany

- Attended lectures by leading researchers on plasma turbulence, kinetic theory, magnetic confinement, magnetohydrodynamics (MHD) and the physics of fusion devices

Molecular Dynamics Research Group

Constructor University

Nov 2025 – Jun 2026

Bremen, Germany

- Researched non-Markovian, non-equilibrium dynamics of aqueous electrolyte systems
- Developed a novel optimisation algorithm based on differential evolution to learn MD memory kernels from simulation data

Image Processing Research Project

Constructor University

Sep 2025 – Dec 2025

Bremen, Germany

- Co-developed a novel SegRAVIR-type U-Net convolutional architecture for semantic segmentation of microvascular structures in infrared retinal fundus images, aimed at early detection of diabetic retinopathy

Nanophotonics Research Assistant

UiT The Arctic University of Norway

Jun 2025 – Aug 2025

Tromsø, Norway

- Independently developed an automation pipeline for an optical-tweezer-coupled Raman spectroscopy system
- Designed and implemented dynamics-based classification of mesoscopic nanoparticle morphotypes

Teaching Experience

Teaching Assistant

Constructor University

Sep 2024 – Jun 2026

Bremen, Germany

- Statistical Physics (Jan 2026 – Jun 2026)
- Scientific Data Analysis; Electrodynamics & Relativity (Aug 2025 – Dec 2025)
- Matrix Algebra & Advanced Calculus I & II (Sep 2024 – Jun 2025)

Leadership Experience

President, Rowing Team

2024 – Jun 2026

Constructor University

Bremen, Germany

- Led training sessions five times a week while keeping the team motivated and engaged
- Led the team to the fastest men's eight (M8+) time in Constructor University's history, currently the fastest student M8+ in Northern Germany

Undergraduate Major Representative, Physics & Data Science

Aug 2024 – Jan 2026

Constructor University

Bremen, Germany

- Represented students on all matters relating to the Physics & Data Science major
- Coordinated a guest lecture and took part in hiring an Associate Professor of Advanced Materials Science
- Raised student concerns with the university's Board of Academic Affairs

Head of Public Relations, Physics Society

Aug 2023 – Jan 2024

Constructor University

Bremen, Germany

- Reached out to professors at other universities to host guest lectures on campus and online
- Planned and promoted a major physics seminar open to all undergraduate students

Skills & Languages

Programming: Python, C++, C#, Julia, Mathematica, SageMath

Markup & Data: LaTeX, HTML, CSS, JSON

Tools: Git, GitHub, Jupyter, Anaconda, VS Code, PyCharm, Arduino

Languages: English (native), Spanish (B2), German (B2), French (A2), Mandarin (A1)